

Parking, Rear and License Plate Lamps



Special Tool(s)

 ST3093-A	Fluke 77-1V Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent

Principles of Operation

The Steering Column Control Module (SCCM) monitors the headlamp switch position by sending voltage signals on multiple circuits to the headlamp switch. There is one circuit for each headlamp switch position. At any given time, one of the signal circuits is switched to ground. The SCCM sends a message to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) , indicating the headlamp switch position.

When the BCM receives a network message from the SCCM that the headlamp switch is in the PARKING LAMPS ON position, the BCM provides voltage to the parking lamps.

If the SCCM does not detect any active inputs from the headlamp switch for 5 seconds, or if the SCCM detects multiple headlamp switch input circuits are active, the SCCM sends a message to the BCM to indicate the fault. The BCM then turns the parking lamps and headlamps on and keeps them on until the battery saver feature times out.

Refer to Exterior Lighting in the Description and Operation portion of this section for information regarding the battery saver feature.

If either situation occurs, the SCCM and the BCM **cannot** be ruled immediately as being at fault. This is normal behavior of the SCCM and the BCM when a fault has been detected with the inputs from the headlamp switch.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection Chart

Mechanical	Electrical
Headlamp switch	- Body Control Module (BCM) fuse(s): 30 (15A) (front parking lamps) 40 (10A) (rear parking lamps) - Bulb - Bulb socket - Exterior mirrors - Wiring, terminals or connectors

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
4. **NOTE:** *Make sure to use the latest scan tool software release.*
- If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .
5. **NOTE:** *The Vehicle Communication Module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM .*
- If the scan tool does not communicate with the VCM :
- check the VCM connection to the vehicle.
 - check the scan tool connection to the VCM .
 - refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.
6. If the scan tool does not communicate with the vehicle:
- verify the ignition is on.
 - The air bag warning indicator prove-out confirms ignition on (other indicators may not prove ignition on).
If the ignition does not turn on, refer to Section 211-05 to diagnose no power in run.
 - verify the scan tool operation with a known good vehicle.
 - refer to Section 418-00, The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.
7. Carry out the network test.
- If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
 - If the network test passes, retrieve and record the continuous memory DTCs.
8. Clear the continuous DTCs and carry out the self-test diagnostics for the BCM .
9. If the DTCs retrieved are related to the concern, refer to Diagnostic Trouble Code (DTC) Chart in this section. For all other DTCs, refer to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.
10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

Symptom Chart

Symptom Chart		
Condition	Possible Causes	Action
One or more parking, rear or license plate lamps is inoperative	- Fuse - Wiring, terminals or connectors	VERIFY the bulbs are OK. If OK, GO to Pinpoint Test O.

	• Bulb socket • Exterior mirror • Body Control Module (BCM)	
• The parking, rear or license plate lamps are on continuously	• Wiring, terminals or connectors • BCM	• GO to Pinpoint Test P.

Pinpoint Tests

Pinpoint Test O: One Or More Parking, Rear or License Plate Lamps Is Inoperative

Refer to Wiring Diagrams Cell [92](#) , Parking, Rear and License Lamps for schematic and connector information.

Normal Operation

The Steering Column Control Module (SCCM) sends voltage signals to the headlamp switch through the headlamp switch input circuits (off, parking lamps, headlamps, autolamps). At any given time, the headlamp switch routes one of the input circuits to ground, indicating the headlamp switch position to the SCCM .

The SCCM sends the status of the headlamp switch position to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) .

The BCM provides voltage to the parking lamps based on the headlamp switch position.

If the SCCM detects multiple or no active headlamp switch inputs, the SCCM sends a message to the BCM to indicate a fault with the headlamp switch input.

If the BCM receives a message from the SCCM indicating a fault with the headlamp switch inputs or the BCM loses communication with the SCCM , the BCM defaults the exterior lamps (headlamps and parking lamps) on.

There are 2 output circuits from the BCM for the parking lamps. One is for the front parking lamps, the other is for the rear parking lamps.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Bulb socket
- Exterior mirror
- BCM

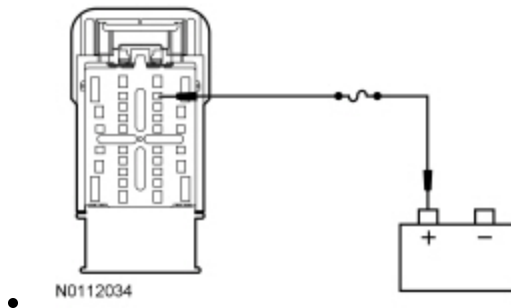
PINPOINT TEST O : ONE OR MORE PARKING, REAR OR LICENSE PLATE LAMPS IS INOPERATIVE	
NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.	
O1 DETERMINE IF ALL THE PARKING LAMPS ARE INOPERATIVE	
• Ignition OFF. • Place the headlamp switch in the PARKING LAMPS ON position.	
Are all the parking lamps inoperative?	

Yes	GO to O7 .
No	If all the rear parking lamps are inoperative, VERIFY the BCM fuse 40 (10A) is OK. If OK, GO to O2. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. If all the front parking lamps are inoperative, VERIFY the BCM fuse 30 (15A) is OK. If OK, GO to O4. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. For any other inoperative lamp, GO to O5.

O2 BYPASS THE BCM (REAR PARKING LAMPS)

NOTE: If the fuse in the jumper wire fails, refer to the Wiring Diagrams manual to identify the possible causes of the circuit short.

- Place the headlamp switch in the OFF position.
- Disconnect: BCM [C2280D](#) .
- Connect a fused jumper wire between the positive battery terminal and the BCM [C2280D](#) Pin 18, circuit CBP40 (YE/GN), harness side.



Do the rear parking lamps illuminate?

Yes	REMOVE the jumper wire. GO to O7 .
No	REMOVE the jumper wire. GO to O3 .

O3 CHECK THE STOPLAMP OPERATION

- Observe the stoplamps while applying the brake pedal.

Are the stoplamps illuminated with the brake pedal applied?

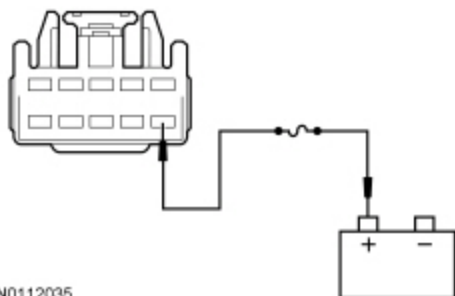
Yes	REPAIR circuit CBP40 (YE/GN) for an open. TEST the system for normal operation.
No	REPAIR circuit GD117 (BK) for an open. TEST the system for normal operation.

O4 BYPASS THE BCM (FRONT LAMPS)

NOTE: If the fuse in the jumper wire fails, refer to the Wiring Diagrams manual to identify the possible causes of the circuit short.

- Place the headlamp switch to the OFF position.

- Disconnect: BCM [C2280E](#) .
- Connect a fused jumper wire between the positive battery terminal and the BCM [C2280E](#) Pin 6, circuit CBP30 (YE/BU), harness side.



Do the front parking lamps illuminate?

Yes	REMOVE the jumper wire. GO to O7 .
No	REMOVE the jumper wire. REPAIR circuit CBP30 (YE/BU) for an open. TEST the system for normal operation.

O5 CHECK FOR VOLTAGE TO THE INOPERATIVE PARKING LAMP

- Place the headlamp switch in the OFF position.
- Disconnect: Inoperative Lamp .
- Place the headlamp switch in the PARKING LAMPS ON position.
- Measure the voltage between the inoperative lamp, harness side and ground as follows:

Inoperative Lamp	Connector-Pin	Circuit
LH front	C1023 Pin 2	CBP30 (YE/BU)
RH front	C1043 Pin 2	CBP30 (YE/BU)
LH front side lamp	C1127 Pin 1	CBP30 (YE/BU)
RH front side lamp	C1126 Pin 1	CBP30 (YE/BU)
LH front marker (Raptor only)	C1596 Pin 1	CBP30 (YE/BU)
RH front marker (Raptor only)	C1597 Pin 1	CBP30 (YE/BU)
LH grille marker (Raptor only)	C1592 Pin 1	CLS30 (VT/WH)
Center grille marker (Raptor only)	C1593 Pin 1	CLS30 (VT/WH)
RH grille marker (Raptor only)	C1594 Pin 1	CLS30 (VT/WH)
High mounted stoplamp (Raptor only)	C904 Pin 5	CBP30 (YE/BU)
LH exterior mirror	C521 Pin 11	CBP30 (YE/BU)
RH exterior mirror	C615 Pin 11	CBP30 (YE/BU)
LH rear upper	C4112 Pin 2	CBP40 (YE/GN)
LH rear lower	C4113 Pin 2	CBP40 (YE/GN)
RH rear upper	C4114 Pin 2	CBP40 (YE/GN)

Inoperative Lamp	Connector-Pin	Circuit
RH rear lower	C4115 Pin 2	CBP40 (YE/GN)
LH rear marker (Raptor only)	C4136 Pin 1	CBP40 (YE/GN)
RH rear marker (Raptor only)	C4138 Pin 1	CBP40 (YE/GN)
LH license plate	C452 Pin 2	CBP40 (YE/GN)
RH license plate	C462 Pin 2	CBP40 (YE/GN)

Is the voltage greater than 10 volts?

Yes	GO to O6 .
No	REPAIR the parking lamp voltage supply circuit in question for an open. TEST the system for normal operation.

O6 CHECK THE INOPERATIVE PARKING LAMP GROUND CIRCUIT FOR AN OPEN

- Measure the voltage between the inoperative lamp, harness side as follows:

Inoperative Lamp	Connector-Pin Circuit	Connector-Pin Circuit
LH front	C1023 Pin 2 CBP30 (YE/BU)	C1023 Pin 1 GD123 (BK/GY)
RH front	C1043 Pin 2 CBP30 (YE/BU)	C1043 Pin 1 GD120 (BK/GN)
LH front side lamp	C1127 Pin 1 CBP30 (YE/BU)	C1127 Pin 2 GD120 (BK/GN)
RH front side lamp	C1126 Pin 1 CBP30 (YE/BU)	C1126 Pin 2 GD123 (BK/GY)
LH front marker (Raptor only)	C1596 Pin 1 CBP30 (YE/BU)	C1596 Pin 2 GD123 (BK/GY)
RH front marker (Raptor only)	C1597 Pin 1 CBP30 (YE/BU)	C1597 Pin 2 GD120 (BK/GN)
LH grille marker (Raptor only)	C1592 Pin 1 CLS30 (VT/WH)	C1592 Pin 2 GD123 (BK/GY)
Center grille marker (Raptor only)	C1593 Pin 1 CLS30 (VT/WH)	C1593 Pin 2 GD123 (BK/GY)
RH grille marker (Raptor only)	C1594 Pin 1	C1594 Pin 2

Inoperative Lamp	Connector-Pin Circuit	Connector-Pin Circuit
	CLS30 (VT/WH)	GD123 (BK/GY)
LH exterior mirror	C521 Pin 11 CBP30 (YE/BU)	C521 Pin 8 GD134 (BK/VT)
RH exterior mirror	C615 Pin 11 CBP30 (YE/BU)	C615 Pin 8 GD139 (BK/YE)
High mounted stoplamp (Raptor only)	C904 Pin 5 CBP30 (YE/BU)	C904 Pin 1 GD139 (BK/YE)
LH rear upper	C4112 Pin 2 CBP40 (YE/GN)	C4112 Pin 1 GD117 (BK)
LH rear lower	C4113 Pin 2 CBP40 (YE/GN)	C4113 Pin 1 GD117 (BK)
RH rear upper	C4114 Pin 2 CBP40 (YE/GN)	C4114 Pin 1 GD117 (BK)
RH rear lower	C4115 Pin 2 CBP40 (YE/GN)	C4115 Pin 1 GD117 (BK)
LH rear marker (Raptor only)	C4136 Pin 1 CBP40 (YE/GN)	C4136 Pin 2 GD117 (BK)
RH rear marker (Raptor only)	C4138 Pin 1 CBP40 (YE/GN)	C4138 Pin 2 GD117 (BK)
LH license plate	C452 Pin 2 CBP40 (YE/GN)	C452 Pin 1 GD117 (BK)
RH license plate	C462 Pin 2 CBP40 (YE/GN)	C462 Pin 1 GD117 (BK)

Is the voltage greater than 10 volts?

Yes	For exterior mirror lamp, INSTALL a new exterior mirror. REFER to Section 501-09. For all other lamps, INSTALL a new bulb socket in question. TEST the system for normal operation.
No	REPAIR the ground circuit in question for an open. TEST the system for normal operation.

O7 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.

- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the **BCM** connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test P: The Parking, Rear Or License Plate Lamps Are On Continuously

Refer to Wiring Diagrams Cell [92](#) , Parking, Rear and License Lamps for schematic and connector information.

Normal Operation

The Steering Column Control Module (SCCM) sends voltage signals to the headlamp switch through the headlamp switch input circuits (off, parking lamps, headlamps, autolamps). At any given time, the headlamp switch routes one of the input circuits to ground, indicating the headlamp switch position to the **SCCM** .

The **SCCM** sends the status of the headlamp switch position to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) .

The **BCM** provides voltage to the parking lamps when the it receives a message from the **SCCM** indicating the headlamp switch in the PARKING LAMPS ON position.

If the **SCCM** detects multiple or no active headlamp switch inputs, the **SCCM** sends a message to the **BCM** to indicate a fault with the headlamp switch input.

If the **BCM** receives a message from the **SCCM** indicating a fault with the headlamp switch inputs or the **BCM** loses communication with the **SCCM** , the **BCM** defaults the exterior lamps on.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- **BCM**

PINPOINT TEST P : THE PARKING, REAR OR LICENSE PLATE LAMPS ARE ON CONTINUOUSLY

P1 CHECK IF THE HEADLAMPS ARE ON

- Ignition ON.
- Observe the headlamps.

Are the headlamps illuminated?

Yes	GO to Pinpoint Test C.
No	GO to P2 .

P2 CHECK THE BCM REAR PARKING LAMP OUTPUT CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: BCM [C2280D](#) .
- Ignition ON.

Do the rear lamps continue to illuminate?

Yes	REPAIR circuit CBP40 (YE/GN) for a short to voltage. TEST the system for normal operation.
No	GO to P3 .

P3 CHECK THE BCM FRONT PARKING LAMP OUTPUT CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: BCM [C2280E](#) .
- Ignition ON.

Do the parking lamps continue to illuminate?

Yes	REPAIR circuit CBP30 (YE/BU) for a short to voltage. TEST the system for normal operation.
No	GO to P4 .

P4 CHECK FOR CORRECT BCM OPERATION

- Ignition OFF.
- Disconnect all the BCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

